№ 3

$$\xi \sim p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \theta > 0$$

Оценки θ: 1.) $\tilde{\theta}_1 = \overline{X}$ 2.) $\tilde{\theta}_2 = X_{(2)}$

а.) Несмещённость оценок:

$$M\xi = \int_{-\infty}^{+\infty} x\, p(x)\, dx = \frac{1}{\theta}\int_0^{+\infty} x e^{-\frac{x}{\theta}}\, dx =$$

$$= \frac{1}{\theta}(-\theta)\int_0^{+\infty} x\, de^{-\frac{x}{\theta}} = -1\left(xe^{-x/\theta}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-\frac{x}{\theta}}\, dx\right) =$$

$$= -\theta e^{-\frac{x}{\theta}}\Big|_0^{+\infty} = \theta$$

1.) $M\tilde{\theta}_1 = M\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\cdot n \cdot M\xi = \theta$

$\Rightarrow \tilde{\theta}_1$ – несмещ.

$$F(x) = \int_{-\infty}^{x} p(t)\, dt = \int_0^{x} e^{-\frac{t}{\theta}}/\theta\, dt =$$

$$= \frac{1}{\theta}(-\theta)\, e^{-t/\theta}\Big|_0^x = e^{-t/\theta}\Big|_x^0 =$$

$$= 1 - e^{-x/\theta} \qquad F(x) = 1 - e^{-\frac{x}{\theta}} \quad \{0; +\infty\}$$

$$\xi_{(2)} \sim \sum_{i=2}^{3} C_3^i F^i(t)(1-F(t))^{3-i} =$$

$$= C_3^2 F^2(t)(1-F(t)) + F^3(t) =$$

$$= \frac{3!}{2!\cdot 1!} F^2(t)(1-F(t)) + F^3(t) =$$

$$= 3F^2(t)(1-F(t)) + F^3(t) =$$

$$= 3F^2(t) - 2F^3(t)$$ ф-ия распр 2-го члена вариационного ряда для выборки объёма $n = 3$.

$$\varphi(t) = 6F(t)\cdot F'(t) - 6F^2(t)F'(t) =$$

$$= 6F'(t)(F(t) - F^2(t)) =$$

$$= \frac{6}{\theta} e^{-x/\theta}(F - F^2) \quad \{0; +\infty\}$$ пл-ть 2-го члена вар. ряда

$$\mu_{X_{(2)}} = \int_0^{+\infty} x \frac{6}{\theta} e^{-\frac{x}{\theta}} (F - F^2)\, dx =$$

$$= \frac{6}{\theta} \int_0^{+\infty} x \cdot e^{-\frac{x}{\theta}} \left(1 - e^{-\frac{x}{\theta}} - (1 - e^{-\frac{x}{\theta}})^2\right) dx =$$

$$= \frac{6}{\theta} \int_0^{+\infty} x \cdot e^{-\frac{x}{\theta}} \left(\cancel{1} - e^{-\frac{x}{\theta}} - (\cancel{1} - 2e^{-\frac{x}{\theta}} + e^{-\frac{2x}{\theta}})\right) dx =$$

$$= \frac{6}{\theta} \int_0^{+\infty} x \cdot e^{-\frac{x}{\theta}} \left(e^{-\frac{x}{\theta}} - e^{-\frac{2x}{\theta}}\right) dx =$$

$$= \frac{6}{\theta} \int_0^{+\infty} x e^{-\frac{2x}{\theta}} dx - \frac{6}{\theta} \int_0^{+\infty} x e^{-\frac{3x}{\theta}} dx =$$

$$= -\frac{\cancel{6}^{3}}{\cancel{\theta}} \frac{\cancel{\theta}}{2} \int_0^{+\infty} x\, de^{-\frac{2x}{\theta}} + \frac{\cancel{6}^{2}}{\cancel{\theta} \cdot 3} \frac{\cancel{\theta}}{} \int_0^{+\infty} x\, de^{-\frac{3x}{\theta}} =$$

$$= -3\left(\cancel{x e^{-\frac{2x}{\theta}} \Big|_0^{+\infty}} - \int_0^{+\infty} e^{-\frac{2x}{\theta}} dx\right) + 2\left(\cancel{x e^{-\frac{3x}{\theta}} \Big|_0^{+\infty}} - \int_0^{+\infty} e^{-\frac{3x}{\theta}} dx\right) =$$

$$= 3 \int_0^{+\infty} e^{-\frac{2x}{\theta}} dx - 2 \int_0^{+\infty} e^{-\frac{3x}{\theta}} dx =$$

$$= -3 \cdot \frac{\theta}{2} e^{-\frac{2x}{\theta}} \Big|_0^{+\infty} + 2 \frac{\theta}{3} e^{-\frac{3x}{\theta}} \Big|_0^{+\infty} =$$

$$= \cancel{-\frac{3\theta}{2} e^{-\frac{2x}{\theta}}} = -\frac{3\theta}{2}(0-1) + \frac{2}{3}\theta(0-1) =$$

$$= \frac{3\theta}{2} - \frac{2}{3}\theta = \frac{5}{6}\theta \neq \theta$$

$\Rightarrow \widetilde{\Theta}_2$ – шмещённая оценка

поправим:

$$\widetilde{\Theta}_2' = \frac{6}{5} X_{(2)}$$

5.) Сравнение эффективности

($\widetilde{\Theta}_a$ эффективнее $\widetilde{\Theta}_b$, если $\forall$ распр из вер. модели $D\widetilde{\Theta}_a \le D\widetilde{\Theta}_b$ и $\exists$ распр.: $D\widetilde{\Theta}_a < D\widetilde{\Theta}_b$)

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2 p(x)\,dx = \int_0^{+\infty} x^2 \frac{1}{\Theta} e^{-x/\Theta}\,dx =$$

$$= \frac{1}{\Theta}\int_0^{+\infty} x^2 e^{-x/\Theta}\,dx = \frac{1}{\Theta}(-\Theta)\int_0^{+\infty} x^2\,d e^{-x/\Theta} =$$

$$= -\left(\cancelto{0}{x^2 e^{-x/\Theta}\Big|_0^{+\infty}} - \int_0^{+\infty} e^{-x/\Theta}\cdot 2\cdot x\cdot dx\right) =$$

$$= 2\int_0^{+\infty} e^{-x/\Theta}\cdot x\cdot dx = 2(-\Theta)\int_0^{+\infty} x\,d e^{-x/\Theta} =$$

$$= -2\Theta\left(\cancelto{0}{x e^{-x/\Theta}\Big|_0^{+\infty}} - \int_0^{+\infty} e^{-x/\Theta}\,dx\right) =$$

$$= 2\Theta(-\Theta)\, e^{-x/\Theta}\Big|_0^{+\infty} = 2\Theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = \theta^2$$

1. $$D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n^2} n D\xi = \frac{\theta^2}{n}$$

2. $$D\tilde{\theta}_2 = D X_{(2)}$$

$$M X_{(2)}^2 = \int_0^{+\infty} x^2 \frac{6}{\theta} e^{-x/\theta} (F - F^2)\, dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-x/\theta} \left(1 - e^{-x/\theta} - (1 - e^{-x/\theta})^2\right) dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-x/\theta} \left(1 - e^{-x/\theta} - (1 - 2e^{-x/\theta} + e^{-2x/\theta})\right) dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-x/\theta} \left(e^{-x/\theta} - e^{-2x/\theta}\right) dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-2x/\theta} dx - \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-\frac{3x}{\theta}} dx =$$

$$= \frac{\overset{3}{\cancel{6}}}{\cancel{\theta}} \cdot \frac{-\cancel{\theta}}{\cancel{2}} \int_0^{+\infty} x^2 \, de^{-\frac{2x}{\theta}} - \frac{\overset{2}{\cancel{6}}}{\cancel{\theta}} \cdot \frac{-\cancel{\theta}}{\cancel{3}} \int_0^{+\infty} x^2 \, de^{-\frac{3x}{\theta}} =$$

$$= -3\left(\cancel{x^2 e^{-\frac{2x}{\theta}}\Big|_0^{+\infty}}^{\,0} - 2\int_0^{+\infty} x e^{-\frac{2x}{\theta}} dx\right) +$$

$$+ 2\left(\cancel{x^2 e^{-\frac{3x}{\theta}}\Big|_0^{+\infty}}^{\,0} - 2\int_0^{+\infty} x e^{-\frac{3x}{\theta}} dx\right) =$$

$$= 6\frac{\theta}{-2}\int_0^{+\infty} x\,de^{-\frac{2x}{\theta}} - 4\frac{\theta}{-3}\int_0^{+\infty} x\,de^{-\frac{3x}{\theta}} =$$

$$= -3\theta\left(\cancel{xe^{-\frac{2x}{\theta}}\Big|_0^{+\infty}}^{\,0} - \int_0^{+\infty} e^{-\frac{2x}{\theta}}dx\right) +$$

$$+ \frac{4}{3}\theta\left(\cancel{xe^{-\frac{3x}{\theta}}\Big|_0^{+\infty}}^{\,0} - \int_0^{+\infty} e^{-\frac{3x}{\theta}}dx\right) =$$

$$= 3\theta\cdot\frac{\theta}{-2}e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} - \frac{4}{3}\theta\frac{\theta}{-3}e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} =$$

$$= +\frac{3}{2}\theta^2 + \frac{4}{9}\theta^2(-1) = \left(\frac{3}{2} - \frac{4}{9}\right)\theta^2 = \frac{19}{18}\theta^2$$

$$D\tilde{\theta}_2 = M x_{(2)}^2 - (M x_{(2)})^2 =$$

$$= \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{38-25}{36}\theta^2 = \frac{13}{36}\theta^2$$

$$D\tilde{\theta}_2' = D\left(\frac{6}{5}x_{(2)}\right) = \frac{36}{25}D x_{(2)} =$$

$$= \frac{\cancel{36}}{25}\frac{13}{\cancel{36}}\theta^2 = \frac{13}{25}\theta^2$$

при $n=3$:

$$D\tilde{\theta}_1 = \frac{\theta^2}{3}$$

$$D\tilde{\theta}_2' = \frac{13}{25}\theta^2 > \frac{\theta^2}{3}$$

$\Longrightarrow \tilde{\theta}_1$ эффективнее чем $\tilde{\theta}_2'$

в.) Для того чтобы использовать Величайшее н-во Крамера-Рао, нужно док-ть:

- Регулярность модели
- Регулярность оценок

Регулярность модели:

$$p(x,\theta) = \frac{1}{\theta} e^{-x/\theta} \{\overbrace{(0;+\infty)}^{A(\theta)}\}$$

непр. дифф при $\theta \neq 0$, $\theta \in (0,+\infty)$ т.к $\theta > 0$

Информация Фишера:

$$I(\theta) = \int_{A(\theta)} \left(\frac{\partial \ln p(x,\theta)}{\partial \theta}\right)^2 p(x,\theta)\,dx \textcircled{=}$$

$$\ln \frac{1}{\theta} e^{-x/\theta} = \ln e^{-x/\theta} - \ln\theta = -\frac{x}{\theta} - \ln\theta$$

$$\textcircled{=} \int_{A(\theta)} \left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2 \frac{1}{\theta} e^{-x/\theta}\,dx =$$

$$= \frac{1}{\theta} \int_{A(\theta)} e^{-x/\theta} \left(\frac{x^2}{\theta^4} - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right) dx =$$

$$= \frac{1}{\theta^3} \int\limits_{A(\theta)} e^{-x/\theta} \left(\frac{x^2}{\theta^2} - \frac{2x}{\theta} + 1 \right) dx =$$

$$= \frac{1}{\theta^3} \int\limits_{A(\theta)} e^{-x/\theta} \frac{x^2}{\theta^2} dx - \frac{1}{\theta^3} \int\limits_{A(\theta)} \frac{2x}{\theta} e^{-x/\theta} dx +$$

$$+ \frac{1}{\theta^3} \int\limits_{A(\theta)} e^{-x/\theta} dx =$$

$$= \frac{1}{\theta^5} (-\theta) \int\limits_{A(\theta)} x^2 de^{-x/\theta} - \frac{2}{\theta^4} (-\theta) \int\limits_{A(\theta)} x de^{-x/\theta} +$$

$$+ \frac{1}{\theta^3} (-\theta) \int\limits_{A(\theta)} de^{-x/\theta}$$

$$= \frac{1}{\theta^4} \cdot 2\theta^2 + \frac{2}{\theta^3} (-\theta) - \frac{1}{\theta^2} e^{-x/\theta} \Big|_0^{+\infty} =$$

$$= \frac{2}{\theta^2} - \frac{2}{\theta^2} - \frac{1}{\theta^2} (0 - 1) = \frac{1}{\theta^2} -$$

– непр. ~~дифф~~ и > 0 на $(0; +\infty)$

Переписываем пр-ных:

$$\int\limits_{A(\theta)} \frac{\partial P(x, \theta)}{\partial \theta} dx =$$

$$= \cancel{\int_{A(\theta)} \frac{1}{\theta}\left(-\frac{1}{\theta}\right) e^{\cdot\theta} dx} =$$

$$= \cancel{-\frac{1}{\theta^2}\int_0^{+\infty} e^{-x/\theta} dx}$$

$$= \int_{A(\theta)} \frac{\partial}{\partial\theta} \frac{e^{-x/\theta}}{\theta} = \int_{A(\theta)} \frac{\frac{x}{\theta^2} e^{-x/\theta} \cdot \theta - e^{-x/\theta} \cdot 1}{\theta^2} dx =$$

$$= \int_0^{+\infty} \frac{x}{\theta^3} e^{-x/\theta} dx - \frac{1}{\theta^2}\int_0^{+\infty} e^{-x/\theta} dx =$$

$$= \cancel{\frac{1}{\theta^3}(-\theta) e^{-x/\theta}\Big|_0^{+\infty}} - \cancel{\frac{1}{\theta^2}(-\theta) e^{-x/\theta}\Big|_0^{+\infty}} =$$

$$= \cancel{\frac{-1}{\theta^2}(0-1)} + \cancel{\frac{1}{\theta}(0-1)}$$

$$= \frac{1}{\theta^3}(-\theta)\int_0^{+\infty} x \, de^{-x/\theta} - \frac{1}{\theta^2}(-\theta) e^{-x/\theta}\Big|_0^{+\infty} =$$

$$= -\frac{1}{\theta^2}\left(\cancel{x e^{-x/\theta}\Big|_0^{+\infty}}^{\,0} - \int_0^{+\infty} e^{-x/\theta} dx\right) + \frac{1}{\theta}(0-1) =$$

$$= \frac{1}{\theta^2}(-\theta) e^{-x/\theta}\Big|_0^{+\infty} + \frac{1}{\theta}(-1) = \frac{1}{\theta} - \frac{1}{\theta} = 0$$

$\Rightarrow$ модель регулярна

Регулярность оценок:

1., 2. $\tilde{\theta}_1$, $\tilde{\theta}_2'$ – несмещённые

$D\tilde{\theta}_1 = \frac{\theta^2}{n}$, $D\tilde{\theta}_2' = \frac{13}{25}\theta^2$ – ограничены на $\forall$ компакте из $[0; +\infty)$

$\Rightarrow$ по доп. условию обе оценки будут регулярными

Нер-во Крамера-Рао:

$\frac{1}{nI(\theta)} = \frac{\theta^2}{n}$ – нижняя грань

$\Rightarrow$ $\tilde{\theta}_1$ – эффективная оценка

при $n = 3$ $\frac{1}{3I(\theta)} = \frac{\theta^2}{3}$ – нижняя грань

$\frac{13}{25}\theta^2 > \frac{\theta^2}{3}$ $\Rightarrow$ $\tilde{\theta}_2'$ менее эффект.

(собственно как доказали ранее)